

Trees are nilrigid

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Abstract

We study cellular automata (CA) on the unoriented k -regular tree T_k , i.e. continuous maps acting on vertex-labelings of T_k which commute with all automorphisms of the tree. We prove that every CA that is asymptotically nilpotent, meaning every configuration converges to the same constant configuration, is nilpotent, meaning each configuration is mapped to that configuration after finite time.

1 Introduction

A classical one-dimensional cellular automaton (or CA) is a continuous function $f : A^{\mathbb{Z}} \rightarrow A^{\mathbb{Z}}$ (where A is a finite set and $A^{\mathbb{Z}}$ is the Cantor space obtained from the product topology), which commutes with the shift map $\sigma(x)_i = x_{i+1}$ [5]. A CA is *nilpotent* if every configuration is mapped to the same σ -fixed point after n steps, i.e.

$$\exists n \in \mathbb{N} : \exists z \in A^{\mathbb{Z}} : \sigma(z) = z \wedge \forall x \in A^{\mathbb{Z}} : f^n(x) = z.$$

This property is well-known for its applications in the computability theory of CA: given the local rule of a CA, nilpotency is undecidable [7] (even if we know a priori that z is an attractive fixed point for f), which provides the basis for the undecidability of many other dynamical properties. For a recent application, we mention the undecidability proof of conjugacy of cellular automata in [6]. A dynamical consequence of this undecidability result is that there is no recursive bound on n in terms of the local rule of the CA f , among nilpotent CA f .

Nilpotency of CA turns out to have several non-trivial dynamical characterizations. For this paper, the relevant one is the theorem from [3, 4] stating that if every point converges to the same fixed point for the dynamics, i.e. the CA is *asymptotically nilpotent*, then the convergence must be uniform, and thus the CA is nilpotent.

Classical cellular automata can be generalized to cellular automata on groups G , as continuous functions $f : A^G \rightarrow A^G$ that commute with G -translations (maps $x \mapsto gx$ defined by $gx_h = x_{g^{-1}h}$). One can talk about nilpotency and asymptotic nilpotency also in this setting, and we refer to the phenomenon that asymptotically nilpotent CA on all full shifts A^G are nilpotent as the *nilrigidity*

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of the group G . Nilrigidity of $\mathbb{Z}^d$ we proved in [9] and in [12] it was generalized in particular to CA on all residually finite solvable groups.

Cellular automata can also be generalized to subshifts. Subshifts are closed subsets $X \subset A^G$ closed under G -translations, and the cellular automata on X are obtained from cellular automata of the full shift that map X to itself. Again, one can naturally talk about nilpotency, asymptotic nilpotency. In particular, we say a subshift is nilrigid if its asymptotically nilpotent cellular automata are nilpotent.

Nilrigidity is known to fail for CA on many subshifts, even ones with rather strong mixing/specification properties [10, 12], but in the setting of full shifts we have no counterexamples on any group. The most interesting open case [12] for this problem is the two-generator free nonabelian group F_2 . This may be surprising, as in the context of symbolic dynamics on groups (including CA on groups), the group F_2 tends to be easier to work with than for example the grid $\mathbb{Z}^2$, because it is geometrically a tree. By results of [12], the group F_2 is nilrigid if and only if all virtually (free non-abelian) groups are nilrigid.

The group F_2 is residually finite, but of course it is not solvable, so the result of [12] does not apply. However, much can be indeed be said about F_2 using the existing methods, in particular we noticed already while writing [12] that these methods can be used to prove nilrigidity for CA that “cannot distinguish between different directions in the tree”. That proof is, however, extremely technical. In the present paper, we give a simple, short and self-contained proof, using only ideas from [3, 4], basic topological dynamics and easy observations about the geometry of trees. Our precise result in terms of CA on group actions is as follows.

Theorem. *Let $k \geq 2$, let T_k be the infinite k -regular (undirected) tree, and let G_k be its full automorphism group. Then every asymptotically nilpotent CA on (any full shift on) the group action $G_k \curvearrowright T_k$ is nilpotent.*

Note that this deals with a generalization of cellular automata in yet another direction, namely continuous self-maps of A^{T_k} that commute with the natural G_k , see Definition 1 below.

This can indeed be interpreted as a result about CA that cannot distinguish between directions in the tree. The above theorem is proved as Theorem 2 below, and we see it as a “near miss” towards the nilrigidity of F_k . It is not clear that our theorem this recovers the result of Guillon and Richard, i.e. nilrigidity of CA on $\mathbb{Z}$, but in the case $k = 2$ we do obtain the special case of nilrigidity for CA on $\mathbb{Z}$ which commute with the mirror map $m(x)_i = x_{-i}$. Neither of the known proofs ([4, 11]) of nilrigidity of $\mathbb{Z}$ seems to be simplified by this assumption. We also do not know a proof of the above theorem which does not rely on the nilrigidity of $\mathbb{Z}$.

For all $k \geq 2$, the action $G_k \curvearrowright T_k$ admits many CA, as any local rule that determines the new symbol at $t \in T_k$ in terms of a finite ball around t (in the path metric), and which is invariant under automorphisms of the ball (as a finite graph) applied to such patterns, defines a CA on $G_k \curvearrowright T_k$. Using conveyor belt methods (see e.g. [1]) it is possible to simulate arbitrary one-dimensional CA this way, and one can show that nilpotency is still undecidable for CA on $G_k \curvearrowright T_k$.

2 Definitions

Definition 1. Let T be a set, $G \curvearrowright T$ a group action. A cellular automaton (CA) on $G \curvearrowright T$ is a continuous function $f : A^T \rightarrow A^T$, where A is a finite set called the alphabet (whose elements are called symbols) and A^T has the product topology, such that $f \circ g = g \circ f$ for all $g \in G$, where the action $G \curvearrowright A^T$ is defined by $gx_t = x_{g^{-1}t}$.

The G -system A^T is called the *full shift* (on the action $G \curvearrowright T$), and elements of A^T are called *configurations*. When T is countably infinite, as a topological space A^T is just the Cantor space, and a standard metric for the topology is $d(x, y) = 2^{-\min\{n | x_{t_n} \neq y_{t_n}\}}$, where $(t_n)_n$ is any enumeration of T .

Remark 1. CA on a group action admit local rules, just like in the classical cases $\mathbb{Z}^d$ and more generally CA on groups [2]. Namely, directly by the definition of continuity, if we pick some $o \in T$ arbitrarily, $f(x)_o$ is determined by some finite restriction $x|_N$, and we can think of N as the neighborhood of o . If the action is transitive, by moving o to an arbitrary $t \in T$, N is translated to a neighborhood for t , and the local rule is also transported this way. Of course, unlike in the case of CA on groups, not every function from A^N to A gives rise to a CA. This point of view is explored in at least [8, 13].

For $k \geq 2$, write T_k for the infinite (rootless) leafless undirected k -regular tree, and G_k for its automorphism group (group of bijections on vertices that preserve the edges), a discrete¹ uncountable group.

One can think of T_k as being defined up to isomorphism, but we give some implementations:

- We can take the vertices V to be finite words over the alphabet $\{0, 1, \dots, k-1\}$, including the empty word, such that $k-1$ can only occur as the first symbol, i.e. the regular language $V = \{\epsilon\} \cup \{0, 1, \dots, k-1\}\{0, 1, \dots, k-2\}^*$, and undirected edges E the sets $\{w, wa\}$ where $w, wa \in V$ and $a \in \{0, 1, \dots, k-1\}$.
- We can take undirected Cayley graph of the free product $\mathbb{Z}_2 * \mathbb{Z}_2 * \dots * \mathbb{Z}_2$ of k copies of the two-element group $\mathbb{Z}_2$, with generators those of the $\mathbb{Z}_2$.
- If k is even, we can use the undirected Cayley graph of the free group $F_{k/2}$ on $k/2$ generators, with the natural generators.

Slightly abusing notation, we write A^{T_k} for the set of all functions $\alpha : V \rightarrow A$ where V is the set of vertices of T_k . By similar abuse of notation, CA on the k -ary tree are CA $f : A^{T_k} \rightarrow A^{T_k}$ on the action $G_k \curvearrowright T_k$.

Definition 2. Let X be a topological space and $f : X \rightarrow X$ a function. We say f is asymptotically nilpotent (towards $0 \in X$) if $f^n(x) \rightarrow 0$ for all $x \in X$. We say f is uniformly asymptotically nilpotent if it is asymptotically nilpotent, and the convergence is uniform over X . We say f is nilpotent (towards $0 \in X$) if there exists $n \in \mathbb{N}$ such that $f^n(x) = 0$ for all $x \in X$. A point $x \in X$ is mortal (towards $0 \in X$) if there exists $n \in \mathbb{N}$ such that $f^n(x) = 0$.

¹This group admits a natural locally compact topology, but this plays no role here.

Definition 3. We say a group action $G \curvearrowright T$ is *nilrigid* if for every finite alphabet A , every asymptotically nilpotent CA $f : A^T \rightarrow A^T$ is uniformly asymptotically nilpotent. A group G is *nilrigid* if the regular action $G \curvearrowright G$ defined by $g \cdot g' = gg'$ is nilrigid.

For $a \in A$ and T a set, write a^T for the unique element of $\{a\}^T$. When studying CA on transitive group actions, any type of nilpotency of $f : A^T \rightarrow A^T$ defines the point $0 \in A^T$ in a shift-invariant way, thus necessarily $0 = 0^T$ for some $0 \in A$. It is easy to see that if $G \curvearrowright T$ is transitive then a CA $f : A^T \rightarrow A^T$ is nilpotent if and only if it is uniformly asymptotically nilpotent.

For $x \in A^{\mathbb{Z}}$ or $x \in A^{\mathbb{N}}$, define $\sigma(x)$ by the formula $\sigma(x)_i = x_{i+1}$. If a symbol $0 \in A$ is clear from context, for any set T and $x, y \in A^T$ such that $\forall t \in T : 0 \in \{x_t, y_t\}$, write $x + y$ for the unique configuration satisfying $\forall t \in T : \{0, (x + y)_t\} = \{x_t, y_t\}$. (This is the cellwise sum of x and y with respect to any abelian group structure on A , or a natural “join” of the two configurations.)

3 Lemmas

We prove some basic lemmas about asymptotic nilpotency. First, configurations must get close to 0 at regular intervals.

Lemma 1. Let X be a compact metrizable space and $f : X \rightarrow X$ a function. If f is asymptotically nilpotent towards 0 then

$$\forall \epsilon > 0 : \exists N \in \mathbb{N} : \forall x \in X : \exists n \leq N : d(f^n(x), 0) < \epsilon.$$

Proof. Suppose not, i.e. there exists $\epsilon > 0$ such that

$$\forall N \in \mathbb{N} : \exists x_N \in X : \forall n \leq N : d(f^n(x_N), 0) \geq \epsilon.$$

Let $x = \lim_i x_{N_i}$ for some sequence $N_1 < N_2 < N_3 < \dots$. Then $d(f^n(x), 0) \geq \epsilon$ for all $n \in \mathbb{N}$. $\square$

Associate to a function $f : A^T \rightarrow A^T$ a bipartite graph G_f where the left and right vertex sets are copies of T , and we have an edge (t, t') from a left vertex to a right vertex if there exist two configurations $x, y \in A^T$ such that $x_u = y_u$ for all $u \neq t$ and $f(x)_{t'} \neq f(y)_{t'}$. Say f has *nice local rules* if the graph G_f is locally finite. This means that $f(x)_t$ only depends on finitely many cells of x , and only finitely many cells of $f(x)$ depend on a particular coordinate x_t .

It is easy to see that continuity of f implies that right vertices have finite neighborhoods in G_f . We note that having nice local rules does not imply continuity when T is infinite: for example if A is totally ordered, then $f(x)_t = \max\{a \in A \mid \{t' \in T \mid x_{t'} = a\} \text{ is infinite}\}$ has nice local rules (the graph G_f has no edges), but is not continuous.

Lemma 2. Suppose $f : A^T \rightarrow A^T$ is continuous and commutes with a transitive group action $G \curvearrowright T$, and the stabilizers of elements $t \in T$ in the action of G have finite orbits. Then f has nice local rules.

Proof. Every $t \in T$ has a unique *minimal neighborhood*, a finite subset $N(t) \subset T$ such that $f(x)_t$ only depends on $x|_{N(t)}$, and $N(t)$ is minimal with this property: some finite set N such that $f(x)_t$ only depends on $x|_N$ exists by continuity, and

the intersection of two neighborhoods is easily seen to be a neighborhood. Since f commutes with every $g \in G$, by the uniqueness of the minimal neighborhood we have $gt = t' \implies gN(t) = N(t')$. It is clear that the graph described in the definition of nice local rules simply records the edges from $N(t)$ to t for all t .

Since f is continuous, the only way for it not to have nice local rules is that there exists some $t \in T$ such that $t \in N(t')$ for infinitely many t' . Since G acts transitively, we can take any such $t' \in T$ and find infinitely many elements of G , say $g_1, g_2, \dots$, such that $t \in N(g_i t') = g_i N(t')$, such that all $g_i t'$ are distinct. By the pigeonhole principle, there is some $t'' \in N(t')$ such that $g_i t'' = t$ for infinitely many g_i , and we restrict to this subsequence. Now observe that for all i, j , the translation $g_i g_j^{-1} \in G$ fixes t , but obviously $g_1 t'$ has infinite orbit under the action of the group $\langle g_i g_j^{-1} \rangle$, meaning the stabilizer of t does not have finite orbits. $\square$

Transitivity of the action does not suffice in the previous statement, as shown by the following example. A similar example was constructed by Törmä (private communication).

Example 1: Consider the set $X_0 \times \mathbb{Z}$ where $X_0 \subset \mathbb{Z}^{\mathbb{N}}$ is the set of sequences with finite support. One can think of this set as a canopy tree with a variable height for the frontier (or variable “camera height”). In this view, the parent of (x, n) is $(\sigma(x), n+1)$ and the children are $(kx, n-1)$, $k \in \mathbb{Z}$. The free group $\langle a, b \rangle$ acts by shifting the “camera”, by $a \cdot (x, n) = (x, n+1)$, and permuting (by $\mathbb{Z}$ -translation) the branch at a fixed level, by $b \cdot (x, n) = (y, n)$ where $y_{-n} = x_{-n} + 1$ if $n \leq 0$ and $y_i = x_i$ for all $i \in \mathbb{N} \setminus \{-n\}$. This action is transitive, and the map $f(\gamma)_{(x,n)} = \gamma_{(\sigma(x), n+1)}$, which copies the value from the parent to the children, commutes with the action of $\langle a, b \rangle$, is continuous, yet does not have nice local rules. $\circ$

We now prove Lemma 3 which says that asymptotic nilpotency implies nilpotency if there are lots of finite mortal configurations. A similar (though strictly speaking incomparable) argument can be found in [4]. The *support* (w.r.t. $0 \in A$) of a configuration $x \in A^T$ is $\text{supp}(x) = \{t \in T \mid x_t \neq 0\}$. We say x is *finite* if its support is.

Lemma 3. *Let T be a set and suppose $f : A^T \rightarrow A^T$ has nice local rules. If f is asymptotically nilpotent towards 0^T and finite mortal configurations are dense in A^T , then f is uniformly asymptotically nilpotent.*

Proof. Suppose f is not uniformly asymptotically nilpotent. Then there exists some $t \in T$ such that for arbitrarily large $k \in \mathbb{N}$, there exists $x \in X$ such that $f^k(x)_t \neq 0$. By Lemma 1, due to asymptotic nilpotency for any finite set $B \subset T$ there exists $N \in \mathbb{N}$ such that for any $x \in A^T$ there exists $n \leq N$ such that $f^n(x)|_B = 0^B$. Combining these two facts, we obtain that for any finite $B \subset T$ there exist arbitrarily large k and configurations $x \in A^T$ such that $x|_B = 0^B$ and $f^k(x)_t \neq 0$.

Since f is asymptotically nilpotent, by the Baire category theorem, for some m there exists an open set U such that every configuration $x \in U$ satisfies $f^k(x)_t = 0$ for all $k \geq m$ (see [4]). Let $y \in U$ be a finite mortal configuration, and suppose $f^{m'}(y) = 0^T$.

Now pick very large finite sets $C \subset B \subset G$ (as a function of the configuration y chosen in the previous paragraph), and choose x as in the first paragraph for

B (and some large k). We claim that if C is large enough and B is large enough compared to C , we have $f^n(x+y) = f^n(x) + f^n(y)$ for all $n \in \mathbb{N}$.

Namely, since y is mortal, if this holds for all $n \leq m'$, it holds for all $n \in \mathbb{N}$. To cover the small n , we pick C so large that every $t \in T$ in the support of any $f^n(y)$, and the minimal neighborhood of any such t , is contained inside C (possible by niceness). Then we pick B so large that the cells in the support of $f^n(x)$, and their minimal neighborhoods, for $n \leq m'$, do not intersect C (possible by niceness). Then there is no interaction between x and y in the first m' steps.

Now, observe that if x is picked as in the previous paragraph so that $f^k(x)_t \neq 0$ for some $k \geq m$, and B is large enough, then we have $x+y \in U$ and $f^k(x+y)_t = f^k(x)_t + f^k(y)_t = f^k(x)_t \neq 0$, a contradiction. $\square$

We say the *support of $x \in A^T$ stays uniformly bounded* if there exists a finite set $S \subset T$ such that $\text{supp}(f^n(x)) \subset S$ for all $n \in \mathbb{N}$. The following lemma is an obvious combination of Lemma 1 and Lemma 3.

Lemma 4. *Let T be a set and suppose $f : A^T \rightarrow A^T$ has nice local rules. If f is asymptotically nilpotent towards 0^T and the support of every finite configuration stays uniformly bounded in the iteration of f , then f is uniformly asymptotically nilpotent.*

Observe that in this lemma, it is not required a priori that the supports of finite configurations stay “uniformly” uniformly bounded in any sense: the bounding set S (and the mortality time) can depend in an arbitrary way on the support of the finite configuration.

We also recall the following result of Guillon and Richard.

Theorem 1 ([4]). *The group $\mathbb{Z}$ is nilrigid.*

4 The theorem

Theorem 2. *The action $G_k \curvearrowright T_k$ of the automorphism group of the k -regular tree T_k is nilrigid.*

Proof. Before giving the formal proof, we give the outline. As in [9] (and also in [12], though less explicitly), a key idea is the use of periodic points, which allow us to apply the strong nilrigidity properties of $\mathbb{Z}$.

In the present case, a natural way to realize this are what we call “stratified configurations” (the set Y defined in detail below), where the symbol in a node only depends on its distance from infinity. Such a configuration is illustrated in Figure 1. It turns out that a CA f on $G_k \curvearrowright T_k$ must preserve the stratified configurations, and in fact acts as a one-dimensional cellular automaton $\bar{f}$ on them.

Now, since f asymptotically nilpotent, $\bar{f}$ is asymptotically nilpotent, and using the nilrigidity of $\mathbb{Z}$, we conclude that $\bar{f}$ is nilpotent, from which we conclude that f is nilpotent on stratified configurations.

Next, we observe that since f commutes with automorphisms of the tree, the support of a finite configuration x can only grow “in a stratified way” out of its geodesic convex hull. More precisely, although f^n may increase the support of x , all nodes from the same distance to the geodesic convex hull of x , and

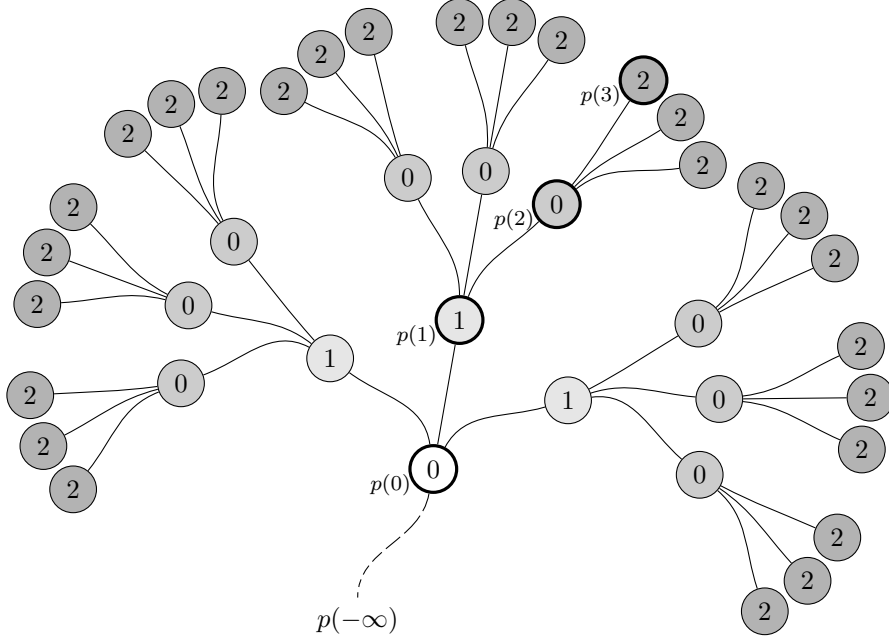


Figure 1: Part of a stratified configuration in the case $k = 4$, using the symbols $A = \{0, 1, 2\}$. The nodes on the path p are indicated with thick circles, and the levels are indicated with fill shades.

in the same branch, must have equal symbols. From nilpotency on stratified configurations, we obtain a bound on how far the support can grow before this local stratification makes it look so close to a point of Y that it must map to 0. This implies that finite configurations cannot grow arbitrarily far, and an application of Lemma 4 concludes the proof.

We now proceed to the formal details. Write $G = G_k$, $T = T_k$. Suppose f is a CA on A^T and $0 \in A$, and $f^n(x) \xrightarrow{n} 0^T$ for all $x \in A^T$. Note that then $f(0^T) = 0^T$ by continuity of f . We need to show that for some $n \in \mathbb{N}$, we have $f^n(x) = 0^T$ for all $x \in A^T$. We denote by $d_T : T \times T \rightarrow \mathbb{R}$ for the path metric in T_k (with values in $\mathbb{N}$). The metric of configurations is d .

Pick a bi-infinite injective path $p : \mathbb{Z} \rightarrow T$, i.e. for all $n \in \mathbb{Z}$, $\{p(n), p(n+1)\}$ is an edge in T and $p(n) \neq p(n+2)$. We stratify T according to distance from “the end $p(-\infty)$ ”, normalized so $p(0)$ is at level 0. More precisely, we define the Busemann function $b : T \rightarrow \mathbb{Z}$ by

$$b(t) = \lim_{n \rightarrow \infty} d_T(p(-n), t) - n.$$

Define $Y \subset A^T$ by $y \in Y \iff \forall t, t' \in T : b(t) = b(t') \implies y_t = y_{t'}$. Thinking of the tree as growing upward from the end $p(-\infty)$, this means nodes on the same height always have the same symbol. A configuration in Y is illustrated in Figure 1.

Define a bijection $\phi : A^{\mathbb{Z}} \rightarrow Y$ by $\phi(x)_t = x_{b(t)}$. We have $f(Y) \subset Y$: if $b(t) = b(t')$ pick a large enough n so the path from $p(-\infty)$ to either each of t or t' contains $p(-n)$. In particular, then the limits defining b have been reached.

Pick an automorphism $g : T \rightarrow T$ that fixes $p(-n')$ for all $n' \geq n$ and maps t to t' . Observe that g fixes every configuration $y \in Y$, so for all $y \in Y$ we have

$$f(y)_t = f(y)_{g^{-1}t'} = gf(y)_{t'} = f(gy)_{t'} = f(y)_{t'}.$$

This is an instance of the general observation that a function commuting with an action cannot make stabilizer subgroups smaller (i.e. cannot break symmetries).

Note that $\phi^{-1}(y)_i = y_{p(i)}$ for $y \in Y$. Observe that $\phi \circ \sigma \circ \phi^{-1} : Y \rightarrow Y$ is induced by any automorphism g of the tree mapping $p(n)$ to $p(n-1)$ for all $n \in \mathbb{Z}$. Similarly, $x \mapsto \phi^{-1}f\phi(x)$ defines a function $\bar{f} : A^{\mathbb{Z}} \rightarrow A^{\mathbb{Z}}$, and because f commutes with g , $\bar{f}$ commutes with $\sigma = \phi^{-1} \circ g \circ \phi$. Since $\bar{f}$ is obviously continuous, it is a CA.

Since f is asymptotically nilpotent, it is asymptotically nilpotent on Y , and it is easy to see that $\bar{f}$ is then asymptotically nilpotent as well. It follows from Theorem 1 that for some $n \in \mathbb{N}$, $\bar{f}^n(x) = 0^{\mathbb{Z}}$ for all $x \in A^{\mathbb{Z}}$. Thus $f^n(y) = 0^T$ for all $y \in Y$.

Clearly f^n is (uniformly) asymptotically nilpotent if and only if f is, so we can w.l.o.g. replace f by f^n . So suppose $f(y) = 0^T$ for all $y \in Y$. It follows that for any $t \in T$, for some $\epsilon > 0$, whenever we have $d(x, y) < \epsilon$ for some $y \in Y$, then $f(x)_t = 0$.

By Lemma 1, it is enough to show that the support of any finite configuration x stays uniformly bounded. It is easy to see that if x and $f(x)$ both have support of size at most one, then their supports are equal. Thus it is enough to show that finite configurations with finite support of size at least 2 stay uniformly bounded. Let thus x be finite with support size at least 2 and $S' \subset T$ its finite support. Let $S \subset T$ be the (geodesic) convex hull of S' , i.e. the smallest set containing S' and the unique shortest path between any pair $s, s' \in S'$.

The idea is now to locally stratify the finite configuration outside S . Far enough outside S , the CA will not be able to distinguish the local stratification from the global one defining Y , and thus its local rule must start writing zero in every cell.

More precisely, consider S as an induced subgraph of T , and let $F \subset S$ be its set of leaves. For each $u \in F$, let B_u be the connected component starting from u outside $S \setminus F$, i.e.

$$B_u = \{t \in (T \setminus S) \cup \{u\} \mid \forall u' \in F \setminus \{u\} : d_T(t, u) < d_T(t, u')\}.$$

Define a map $c_u = c : B_u \rightarrow \mathbb{N}$ by $c(u) = 0$ and in general $c(t) = d_T(t, u)$.

Let Z_u be the set of configurations z where the symbols in the component B_u only depend on the value of $c(t)$, i.e.

$$z \in Z_u \iff (\forall t, t' \in B_u : c(t) = c(t') \implies z_t = z_{t'}).$$

Let $Z = \bigcap_{u \in F} Z_u$. We claim that $f(Z) \subset Z$. For this, it is enough to show $f(Z_u) \subset Z_u$ for all $u \in F$. For this, observe that the subgroup of G_k that pointwise stabilizes $T \setminus B_u$ acts transitively on $c^{-1}(n)$ for all $n \geq 0$. Thus $f(Z_u) \subset Z_u$ is proved by a similar formula as $f(Y) \subset Y$ above.

Suppose then $z \in Z$, and fix an arbitrary $s \in S$. We claim that for some n' , whenever $d_T(t, s) \geq n'$, then $f(z)_t = 0$. To see this, it is enough to show that for all $u \in F$ there exists n such that whenever $z \in Z_u$, $t \in B_u$ and $d_T(t, u) \geq n$, we have $f(z)_t = 0$. Pick any n such that the minimal neighborhood $N(s)$ of s is

contained in the n -ball $\mathcal{B}_n(s) = \{s' \in T \mid d_T(s, s') \leq n\}$ around s . Observe that then $N(t) \subset \mathcal{B}_n(t)$ for all $t \in T$ because the path metric is invariant under G .

Suppose now $z \in Z_u, u \in F, d_T(t, u) = m \geq n, t \in B_u$. Consider any automorphism $g \in G_k$ that maps u to $p(-m)$ and t to $p(0)$, and in general if $t' \in B_u$ and $c(t') = i$, maps t' into $b^{-1}(i - m)$. Such an automorphism exists because B_u is just a $(k-1)$ -regular rooted tree, and thus isomorphic, as a rooted tree, to the connected component of $T_k \setminus p(-m-1)$ rooted at $p(-m)$.

We can find a configuration $y \in Y$ which agrees with gz in every cell at distance at most m from $p(0)$: for any $q : (-\infty, -m-1] \rightarrow A$ there is a unique configuration $y \in Y$ that satisfies $y|_{gB_u} = gz|_{gB_u}$ and $y|_{p(i)} = q(i)$ for $i < -m$. In particular, since $m \geq n$, we have $y|_{\mathcal{B}_n(p(0))} = gz|_{\mathcal{B}_n(p(0))}$, thus $f(gz)_{p(0)} = f(y)_{p(0)} = 0$, thus by commutation with g we have

$$f(z)_t = f(z)_{g^{-1}p(0)} = f(gz)_{p(0)} = 0.$$

It follows that indeed for some n' , whenever $z \in Z$ and $d_T(t, s_0) \geq n'$, we have $f(z)_t = 0$. In other words, $z \in Z$ implies that after an application of f , every nonzero symbol is at distance at most n' from s , and we still have $f(z) \in Z$. Now, if the support of x is originally contained in S , then obviously $x \in Z$, since we have $c_u^{-1}(0) = \{u\}$, and $x_t = 0$ for $t \in B_u \setminus \{u\}$. Thus, for all k , the support of $f^k(x)$ is contained in $\mathcal{B}_{n'}(s_0)$. The uniform asymptotic nilpotency of f now follows from Lemma 4. $\square$

5 Questions

The stabilizers of the action of G_k on T_k have finite but unbounded orbits, so the action is far from simply transitive. The infinite k -regular tree is a Cayley graph for $k \geq 2$, so simply transitive subactions exist: for any k , it is the Cayley graph of k copies of the two-element group $\mathbb{Z}_2$, and for even k , it is the Cayley graph of the free group $F_{k/2}$ with respect to a free generating set of size $k/2$.

Question 1. *Does T_k admit a simply transitive nilrigid group action by graph automorphisms? More generally, for which transitive subgroups $H \leq G_k$ is $H \curvearrowright T_k$ nilrigid?*

The related question of nilrigidity F_k was asked in [12] (the more general question of whether CA on all groups are nilrigid was asked in [9]). Our main result can be seen as a near miss towards this question (because as discussed above, F_k 's action on itself is a particular action on a tree), and Question 1 suggests also a quantitative way to attack this – can we find nilrigid transitive actions on the tree with smaller stabilizers? We also repeat [12, Question 11.8]:

Question 2. *Is every transitive group action nilrigid?*

One can also formulate the question for monoid actions (the opposite monoid acts naturally on configurations). We do not know whether all transitive monoid actions are nilrigid.

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